

IT REQUIREMENTS FOR STORE CONSTRUCTION

Version 2.1

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1. Overview

The requirements set forth in this document apply for any type of store, regardless of size, architecture or location. During the planning and construction phases a close coordination between the Construction, Procurement, Facility Management, Sales, and IT departments is required to ensure smooth store construction process.

Note: Local codes supersede this specification, however any deviation from the requirements must be documented and approved by Store Infrastructure team.

2. Location of IT Cabinets

In the store there are only two suitable locations where the IT cabinets can be installed.

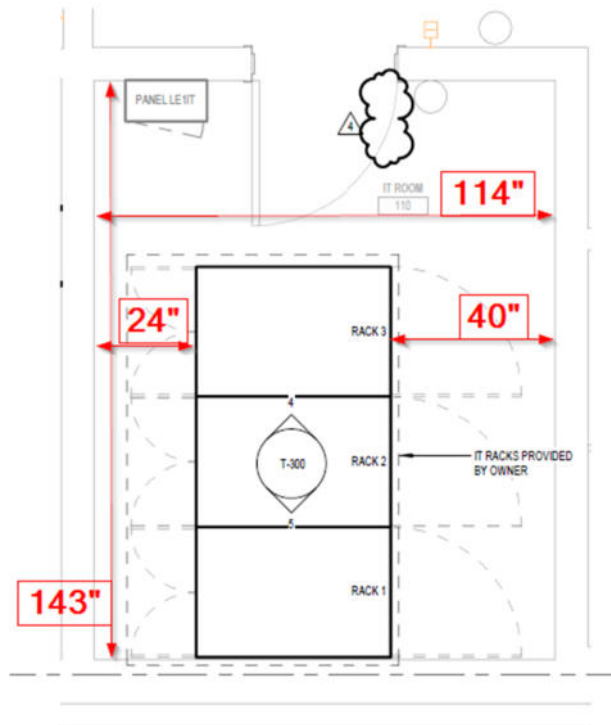
- Server Room (Preferred – Best Option)
- Store Manager's Office

Any deviation for the placements of the cabinets must be agreed in advance with IT.

Due to the multiple types of stores, and their specific characteristics it is not possible to standardize a unique placement layout which covers all the stores, and for this reason it is crucial the IT department in conjunction with the construction and sales departments define the best location for the installation of the cabinets.

3. Server Room

The constructional constraints for the IT technical room are accurately defined in the Prototype Design and must be followed.



- **Minimal** size of the room: 114" x 143" (9.5 ft x 11.9167)
- Minimum height of the ceiling: 10'
- A clearance of 40" in front and 24" behind the racks is mandatory and **cannot be reduced by any means.**
- The corridor from the door to the opposite wall must be a clear and without any obstacles (e.g. racks, fire extinguishers, etc.)
- No suspended ceiling, no raised floor.
- Floor of fire resistance class B1 or higher, without tiles, with antistatic paint.
- Room must be F90 / EI90 fire resistant
 - *Walls* on fire resistance class F90 (no wallpaper, no windows).
 - *Door* minimum 1000 mm width fire rated T60-RS (smoke tight) door.
- Access is secured by a cylinder lock.
- No water pipes, power, duct work or data lines, which are not intended for the operation of the components of the IT room are allowed to pass through it.

- IT components or racks must not be installed beneath the AC units to prevent the consequences of a water leakage.

3.1 Server Room Air Conditioning

- Two wall or ceiling mounted split air-conditioning units, 100% redundant, each with a cooling capacity of 3.5 kW, must be provided.
- The ambient temperature of the room must be maintained constant in the 64°F – 80°F range. Each country is responsible for the most efficient operation of the cooling system according to its specific climate conditions.
- Temperature alarm thresholds: min 50°F - max 95°F (outside of this range the performance of the IT equipment could be degraded, and its life time reduced.)
- The humidity inside the room should remain between 20% and 60% (the installation of dehumidifiers/humidifiers is not necessary).
- Water collection pans must be installed under the AC units.
- Temperature control: An ambient temperature indicator connected to the building management system (GLT) must be installed in the IT room.
- Leakage detector: On the floor of the IT Technical room a leakage detector with connection to the burglar alarm system must be installed directly under the AC splitter units.

3.2 Server Room Alarm systems

- The intrusion alarm (IA) and fire alarm (FA) panels must never be installed inside the IT technical room.
- The installation should be done so the alarm is protected against sabotage and located in an area which is not accessible by public. Local laws and regulations shall be taken in to account. Accessibility for fire fighters and security services needs to be granted.

3.3 CCTV in the Server room

- A camera inside the IT room must be installed and it must be pointed towards the entrance.
- Both the lighting conditions and the resolution of the camera shall permit that the faces recorded by the camera can be easily recognizable.

3.4 Fire Protection of the Server room

- 2 x CO2 fire extinguishers must be placed at the IT technical room, one inside and another one outside the room. Fire extinguishers of different types (e.g. ABC powder or water hoses) must not be placed nearby the IT technical room.
- A smoke detector with connection to the fire alarm system must be installed.

- Country specific regulations must be taken into consideration, especially regarding the weight of the fire extinguishers (5Kg, 2Kg, etc.)
- The cabling entrances to the IT room must be sealed and fire resistant, the use of F90 foam is allowed but recommended methods are: ROXTEC or EZ-PATH, because these allow future growth and do not require to certify the room again when additional cabling must be installed.

4. Cabinet Requisites

Each cabinet shall be accessible from both sides (front/rear) which means there must be a clearance free of fixed obstacles of 3' in front of the cabinet and 2' clearance on the rear. This clearance cannot be reduced for any reason.

5. IT Cabinet Power Requirements

Each IT cabinet must have four (4) Twist-Lock receptacles – two (2) UPS and two (2) utility power.

- Cabinet 1 – Two (2) L5-20R UPS and two (2) L5-20R utility power. The power receptacles shall be clearly labeled with panelboard and circuit number I.D.
- Cabinet 2 – Two (2) L5-30R UPS and two (2) L5-30R utility power. The power receptacles shall be clearly labeled with panelboard and circuit number I.D.
- Cabinet 3 (future) – Two (2) L5-30R UPS and two (2) L5-30R utility power. The power receptacles shall be clearly labeled with panelboard and circuit number I.D.

6. IT Cabinet Grounding Requirements

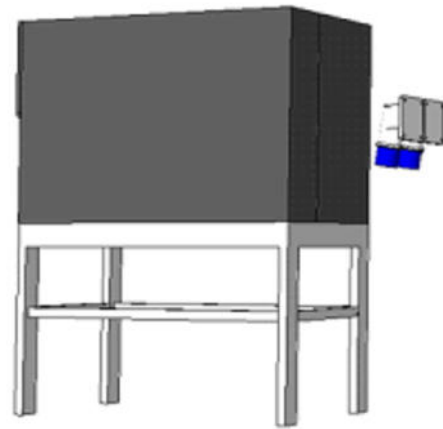
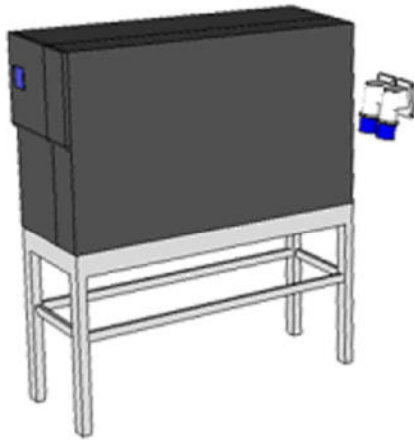
Each IT cabinet must have a separate #6 ground wire from buss bar. Do not daisy chain ground wires.

7. UPS

The installation of a UPS is mandatory. UPS shall be installed by the electrical contractor. Only IT equipment, BMS panel, and specified security equipment shall be connected to the UPS. All UPS power receptacles including twist-locks shall be blue in color.

8. UPS Installation Requirements

- UPS must be installed in close proximity to the LE1IT panel.
- UPS must be installed on a pedestal or support of at least 1' AFF.



- Provide data box with 1" conduit, pull string, and insulated bushing from UPS location to the nearest cable tray. The data box must be mounted within twenty-four inches from the rear of the UPS.
- The UPS unit shall be monitored permanently by IT.

9. Passive Cabling

All data cabling for data, WAP's and CCTV must be in conduit.

- Minimum conduit size 1".
- All conduits require insulated bushings.
- All conduits must be connected to the cable tray using manufacture clips.
- Each conduit must have a pull string installed.
- All boxes must have single gang plaster rings.
- All data boxes must be numbered on plans (D-XX) following the prototype numbering scheme.
- All WAP boxes must be numbered on plans (WAP-XX) following the prototype numbering scheme.
- All CCTV boxes must be numbered on plans (C-XX) following the prototype numbering scheme.
- Data boxes must not be connected together.
- WAP and CCTV boxes may be connected together but shall not exceed the 40% fill rate (cable is .265 in O.D.).
- Conduits must enter and exit the boxes straight in and out and no bends in conduit are allowed when boxes are connected together.

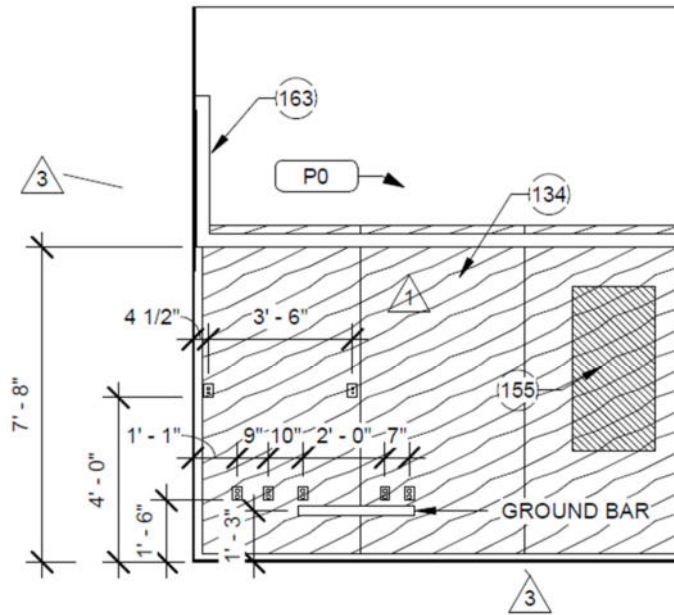
10. Cable Trays

When considering the new cabling routes for data, the specific geometry of each store must be accounted for. Also, the design for these routes must be agreed with the construction department during the planning phase. The shortest possible routes must be taken. When determining the size of the cable trays, the quantity, type, and bend radius requirements of the cable shall be considered. Additionally, an allowance of 50% for growth shall be provided.

- It is strictly prohibited to locate the trays in elevators shafts, as well as close to gears which could cause electromagnetic noise.
- Any penetrations through fire walls shall be done using four-inch EZ Paths. The quantity of the EZ Paths should allow the maximum number of cables to pass through.
- If cable tray requires an elevation change, the tray must be installed using a waterfall at the wall (90 degree outside bend) and a 90 degree inside bend at the lower elevation change.
- All cable tray must be accessible and not blocked with duct work, light fixtures, etc.
- All metallic trays shall be grounded #6 AWG CU, bare grounding conductor. Bond each section of cable tray at two locations. All cable trays must be installed per manufacturers recommendation
- Any water lines / refrigeration piping should be installed below (and not above) the cable tray.

11. Technical Room

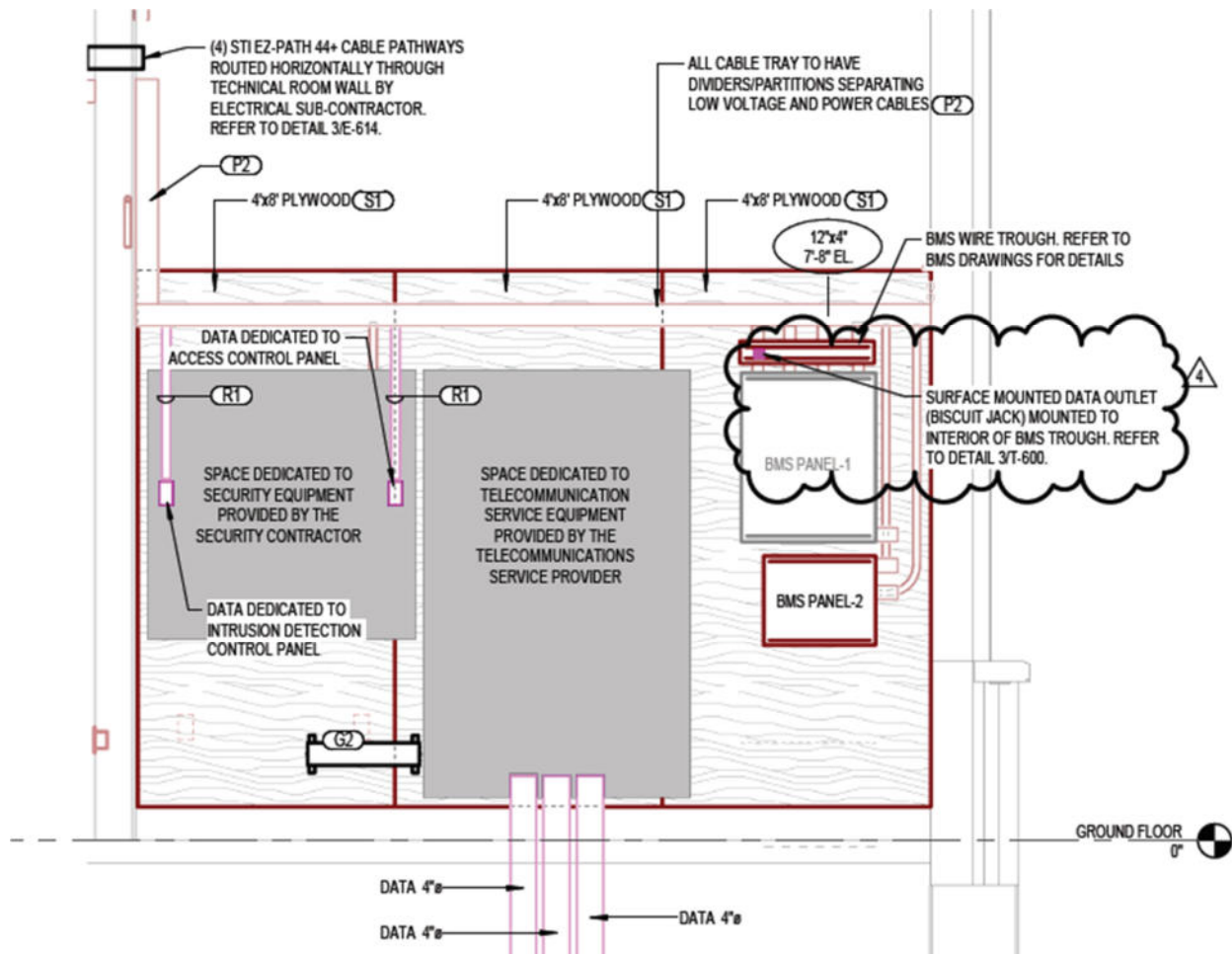
- Provide 4'x8' plywood backboard for Demarc. Plywood backboard shall be AC grade fire resistant plywood with the c-face of the plywood mounted facing the wall. If the plywood backboard is painted, the fire resistant labels shall remain visible. This space shall be dedicated for telecommunication service equipment provided by the telecommunications service provider.
- Provide three 4" conduits with mule tape for service provider entrance directly under the dedicated space from note above.
- Provide ground bar.
- Provide quad UPS receptacle mounted on plywood backboard at 18" AFF. Power receptacle shall be clearly labeled with panelboard and circuit# I.D.
- Provide cable tray and EZ Paths for routing cable into IT Room



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TECHNICAL ROOM ELEVATION

1/4" = 1'-0"



12. Fire Alarm Control Panel

Provide 1"conduit with pull string and insulated bushing connected from the top of the FACP to the nearest cable tray.

13. BMS Panel

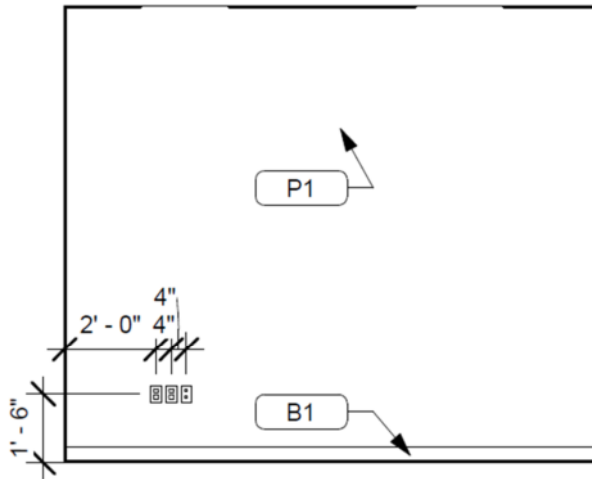
Provide data box with 1"conduit, pull string and insulated bushing from BMS panel location to the nearest cable tray. Data box must be mounted within 6" from the top right corner of panel.

14. XWEB Panel

Provide data box with 1"conduit, pull string and insulated bushing from XWEB panel location to the nearest cable tray. Data box must be mounted within 6" from the top right corner of panel.

15. Store Manager Room

This office requires one data outlet, one UPS duplex power receptacle and one Utility duplex power receptacle no more than six inches apart. The power receptacles shall be clearly labeled with panelboard and circuit# I.D.

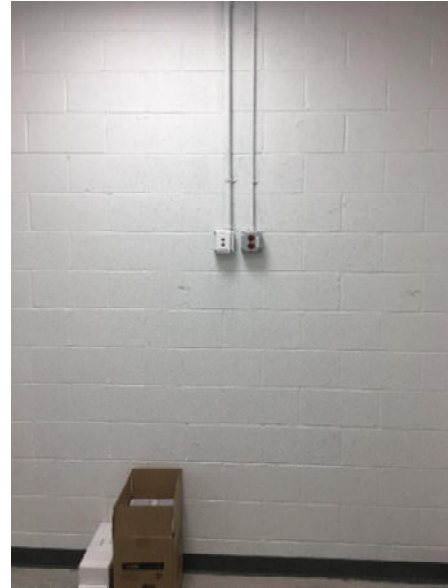
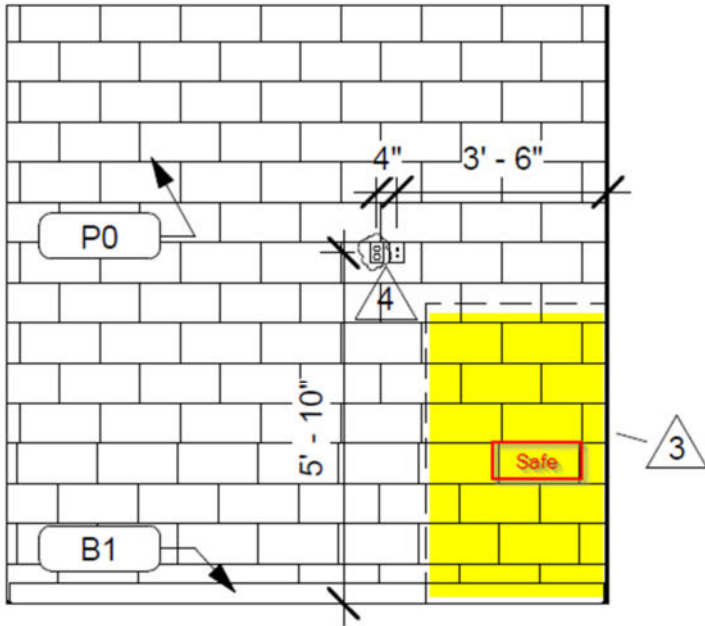


OFFICE 108 ELEVATION

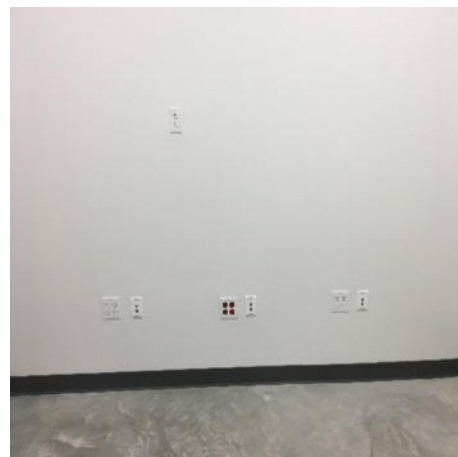
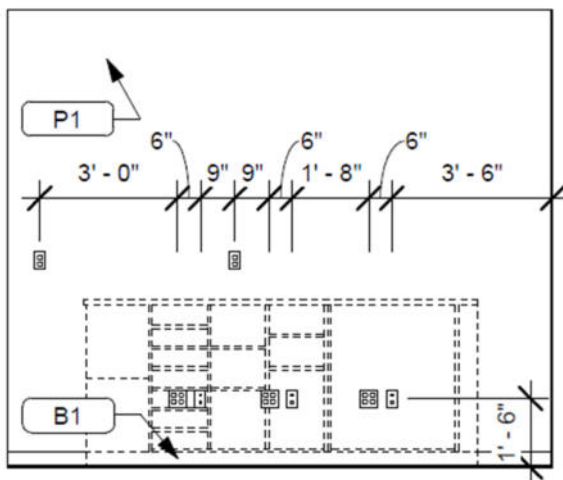
16. Cash Office

This office requires four (4) data outlets , one (1) UPS quad power receptacle and one (1) UPS duplex receptacle.

- One data outlet is for the Safe and it will be mounted on the opposite wall of the room's door. IT will be mounted above the safe at 70" AFF along with one UPS duplex power receptacle no more than 6" apart. The UPS duplex power receptacle shall be clearly labeled with panelboard and circuit number I.D.

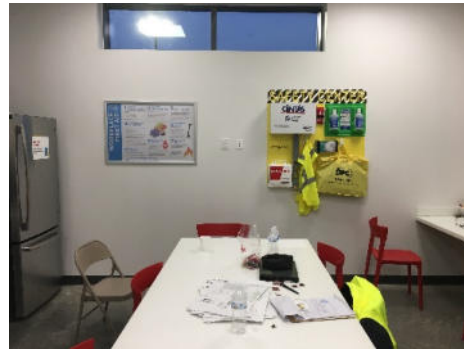
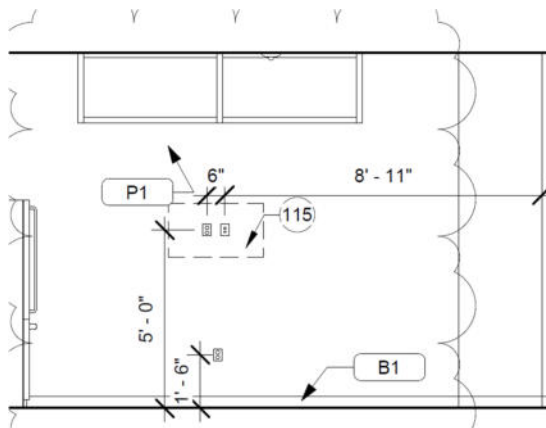
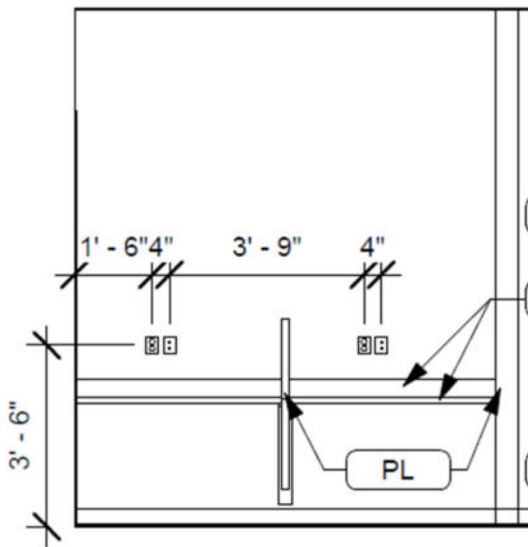


- Three data outlets will be mounted under the countertop 18" AFF with receptacles no more than six inches apart. The two outside power receptacles are utility and the middle power receptacle is a quad UPS. All power receptacles shall be clearly labeled with panelboard and circuit# I.D.



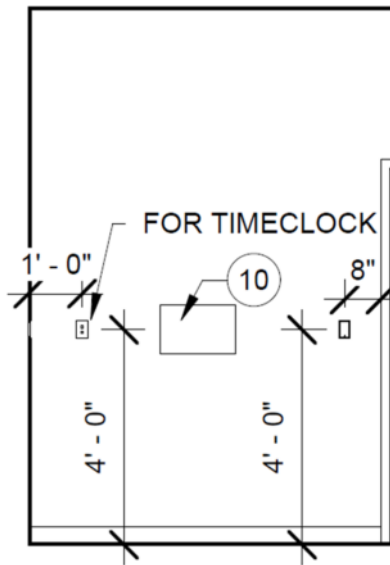
17. Break Room

This area requires three data outlets. One data outlet at each training station (2) at 42" AFF. One data outlet for future display monitor location at 60" AFF. Each data drop requires utility duplex power receptacle which shall be clearly labeled with panelboard and circuit# I.D.



18. Break Room Hallway

- This area requires one data outlet mounted at 48" AFF for the Time Clock.
- One WAP shall be mounted 12" ACT in the center of the hallway.



19. Back of House (Backstock, Staging, Loading Dock)

- WAP and CCTV boxes are installed in various locations in the back of house and require conduits with pull string to the nearest cable tray. See section 27 access points and 28 CCTV for requirements.
- Three WAP boxes are mounted one each on the top of the cooler, freezer and bakery freezer. Each WAP shall be centered on the top.

20. Sales Area

- Cable tray shall be installed above the checkouts. See section 10 Cable trays for requirements. Each checkout requires a power pole installed from checkout fixture and attached to the cable tray. If power pole is not able to be connected to the tray due to fixture alignment, then one ¾" flex (two ¾" for the self-checkouts) must be installed from the cable tray into the top of the power poles with pull string.
- WAP and CCTV boxes are installed in various locations in the sales area and require conduits with pull string to the nearest cable tray. See section 27 access points and 28 CCTV for requirements.
- Display monitor data boxes at front of store must be mounted on the side of the column opposite the store vestibule entrance side. Do not mount on the front of the column.



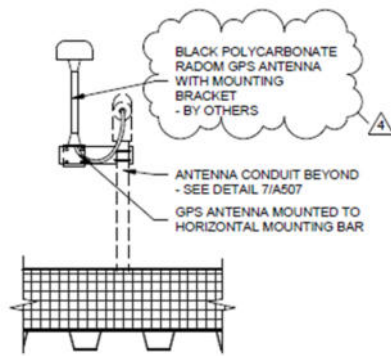
21. Bakery

The Bakery has five data outlets and one GPS conduit.

- Three data outlets are mounted centered directly behind the ovens at 57" AFF.
- One data outlet for future display monitor mounted at 57" AFF above the bakery prep table.
- One data outlet inside of bakery millwork at 48" AFF. This data drop requires utility duplex power receptacle which shall be clearly labeled with panelboard and circuit# I.D. This data drop must be within 6" of the conduit shown in note below.

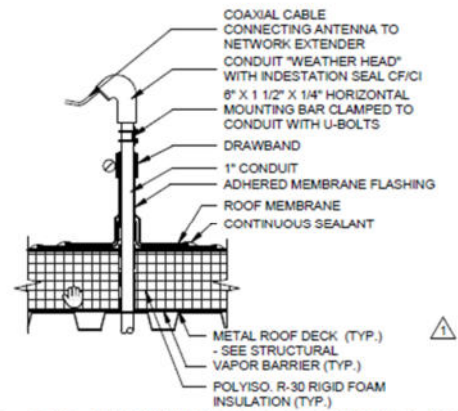
22. GPS Antenna conduit

- The conduit run must not exceed 90' total.
- Mount one junction box inside of bakery millwork at 48" AFF within 6" of the data box shown in note above.
- Install 1" conduit and pull string from the junction box to roof. Cap with weatherhead 18" above roof line for GPS antenna.
- The GPS antenna must have a direct view of GPS satellites with an unobstructed line of sight to the sky (do not mount near walls or any equipment mounted on roof).



GPS ANTENNA MOUNTING DETAIL (TYP.)

1 1/2" = 1'-0"



7 GPS ANTENNA PENETRATION (TYP.)

1 1/2" = 1'-0"



23. SDR

The SDR shall be mounted in the back of house near receiving dock mounted by electrical contractor six inches below the cable tray. Provide two four-inch chase nipples between cable tray and connect into top of the SDR. Any deviation for the placements of the SDR must be agreed in advance with IT. The minimum clearance between floor and bottom of the rack is 8' as specified on prototype drawing A-202-5.



24. SDR Power Requirements

SDR must have two (2) twist-lock receptacles – one (1) L5-20R UPS and one (1) L5-20R utility power mounted inside bottom rear of the cabinet. The power receptacles shall be clearly labeled with panelboard and circuit number I.D.

25. SDR Grounding Requirements

SDR must have a separate #6 ground wire from buss bar.

26. Access Points

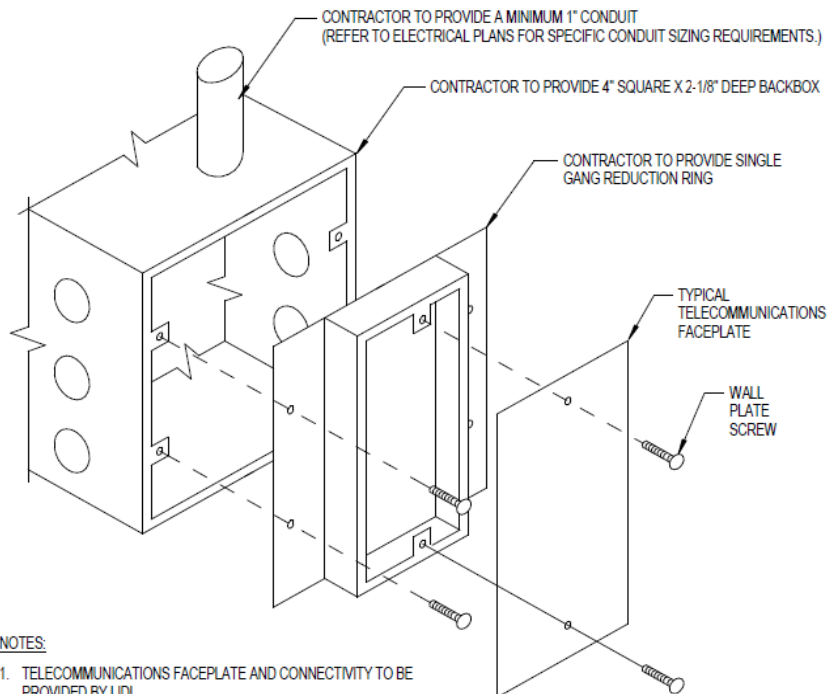
Provide junction box with plaster ring for Wi-Fi data point. Ceiling mounted with 1" conduit to nearest cable tray with pull string and bushing provide vertical conduit down from box to 14' 0" off. Wi-Fi data box needs to be mounted with clear line of sight to the floor do not mount above light fixtures, conduits, duct work, display fixtures, etc. Do not mount within 3' of speakers. See section 9 passive cabling for conduit requirements.

27. CCTV

Provide junction box with plaster ring for CCTV point. Ceiling mounted with 1" conduit to nearest cable tray with pull string and bushing provide. CCTV box needs to be mounted with clear line of sight to the floor do not mount above light fixtures, conduits, duct work, display fixtures, etc. See section 9 passive cabling for conduit requirements.

Cash Office	1 data box above safe location with UPS receptacle - Mount at 5'10"AFF 1 data box below countertop with quad UPS receptacle 1 data box below countertop with utility receptacle 1 data box at printer location with utility receptacle 1 camera box ceiling mounted
Amenity Hallway	1 data box outside cash office for time clock - Mount at 4'AFF 1 data box 12"ACT for WAP
Managers Office	1 data box at desk with UPS and utility receptacles
Break Room	1 data box at loop monitor location with utility receptacle - Mount at 5'AFF 2 data boxes above countertop each with utility receptacle - Mount at 3'6"AFF
FACP	1" conduit from FACP to cable tray for POTS Lines
BMS Panel	1 data box mouthed 6" above top of BMS Panel
Security Panel	1 data box mounted next to Security Panel
Card Access Panel	1 data box mounted next to Card Access Panel
UPS	1 data box mounted 6" above UPS location
XWEB	1 data box mouthed 6" above top of XWEB Panel
Bakery	3 data boxes at ovens each oven has 1 data box centered behind it - Mount at 4'9"AFF 1 data box tile wall - Mount at 4'9"AFF 1 data box inside millwork with utility power must be within 12" of GPS conduit to roof 1 data box location TBD for Deli Scale 1 one inch conduit inside millwork up to roof for GPS antenna must be within 12" of data box in same millwork
Sales Floor	2 data boxes I and each digital signage monitor location with utility power - Mount at 8'2"AFF
WAP's	WAP's shall not be mounted higher than 14' AFF.
Bottle Recycling	2-3 data boxes I and each Bottle Recycling location with utility power (not in all stores)
IT Room	1 camera box mounted on wall Mount at 8'AFF

TYPICAL FLUSH MOUNTED TELECOMMUNICATIONS DEVICE ROUGH-IN DETAIL



NOTES:

1. TELECOMMUNICATIONS FACEPLATE AND CONNECTIVITY TO BE PROVIDED BY LIDL.
2. ROUTE A CONDUIT BETWEEN THE TELECOMMUNICATIONS DEVICE BACK-BOX AND THE NEAREST TELECOMMUNICATION ROOM OR ACCESSIBLE PATHWAY.
3. REFER TO TECHNOLOGY PLANS FOR DEVICE LOCATIONS AND QUANTITIES, CABLE ROUTING AND DESTINATIONS.
4. PROVIDE A PULL STRING ROUTED THROUGH EACH TELECOMMUNICATIONS CONDUIT.
5. PROVIDE A PLASTIC CONDUIT BUSHING AT THE ACCESSIBLE CABLE PATHWAY END OF EACH CONDUIT.
6. PAINT ALL EXPOSED CONDUIT THE SAME AS THE CEILING UNLESS DIRECTED OTHERWISE BY THE OWNER'S REPRESENTATIVE OR ARCHITECT.